

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	ARTHI A	Reg. No.:	192525152
Date:	27.07.2026	Duration:	60 minutes

Code of Conduct

I ARTHI A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Arthi. A

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25

CASE STUDY 36 — SDG 9

Smart Construction Project Management System

AGILE SPRINT PLANNING & USER STORY WORKSHOP

College Name: SIMATS Engineering

Department: Bachelor of Technology in Artificial intelligence & Machine learning

Course Name: Software Engineering

Case Study Title: SDG 9 Smart Construction Project Management System

Student Name: Arthi A

Register Number: 192525152

Faculty Name: Sharmili K

Date of Submission: 27.07.2026

Live Jira prototype: <https://arthia9393-1785414127653.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog?jql=parent+%3D+SCRUM-7&atlOrigin=eyJpIjoiOTVkyY2U3M2M3YmYwNDVIM2JhN2U2ZDYxM2M4YmNIMTMiLCJwIjoiYiJ>

1. Project Vision & Stakeholders

Vision Statement To build a centralized, cloud-based Smart Construction Project Management System that eliminates spreadsheet-driven, manual reporting by unifying project planning, resource/material management, and real-time progress tracking — reducing delays, cost overruns, and communication gaps across construction projects.

Objectives

- Replace manual/spreadsheet tracking with a single centralized platform
- Enable real-time visibility of site progress for all stakeholders
- Streamline material procurement to reduce delays
- Improve communication between field and office roles
- Support data-driven decisions on schedule, cost, and resource allocation

Key Stakeholders

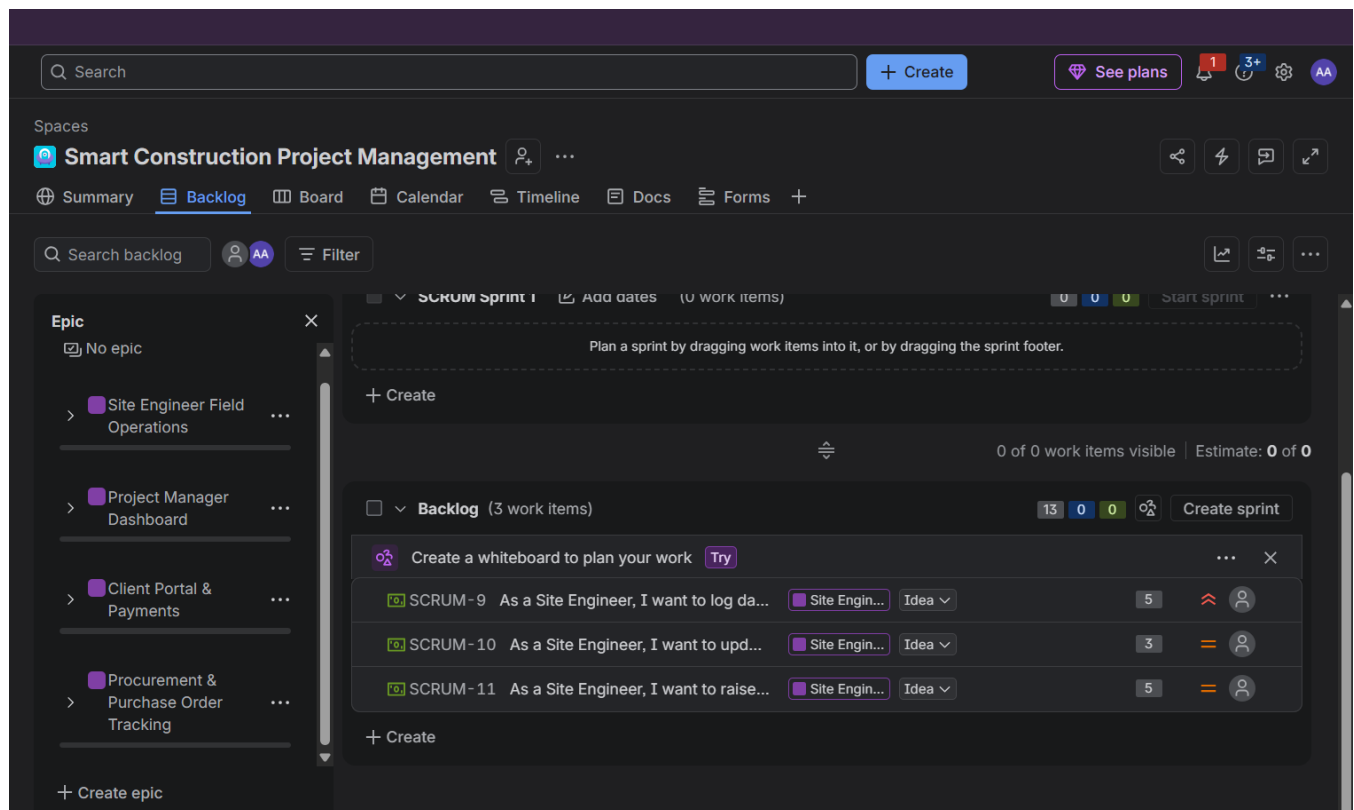
Stakeholder	Role in the System
Site Engineer	Logs daily progress, raises material requisitions, updates task status from the field (mobile)
Project Manager	Tracks schedules, reviews/approves requisitions and milestones, monitors project health (web dashboard)
Procurement Team	Converts approved requisitions into purchase orders, tracks delivery status
Client	Views milestone progress, approves payments tied to completed milestones (client portal)
Company Admin/Owner	Oversees multiple projects, resource allocation across sites
Suppliers/Vendors	Fulfill purchase orders raised by procurement (indirect stakeholder)

2. Epics & User Stories

(4 Epics, each mapped 1:1 to a Figma screen)

Epic 1: Site Engineer Field Operations

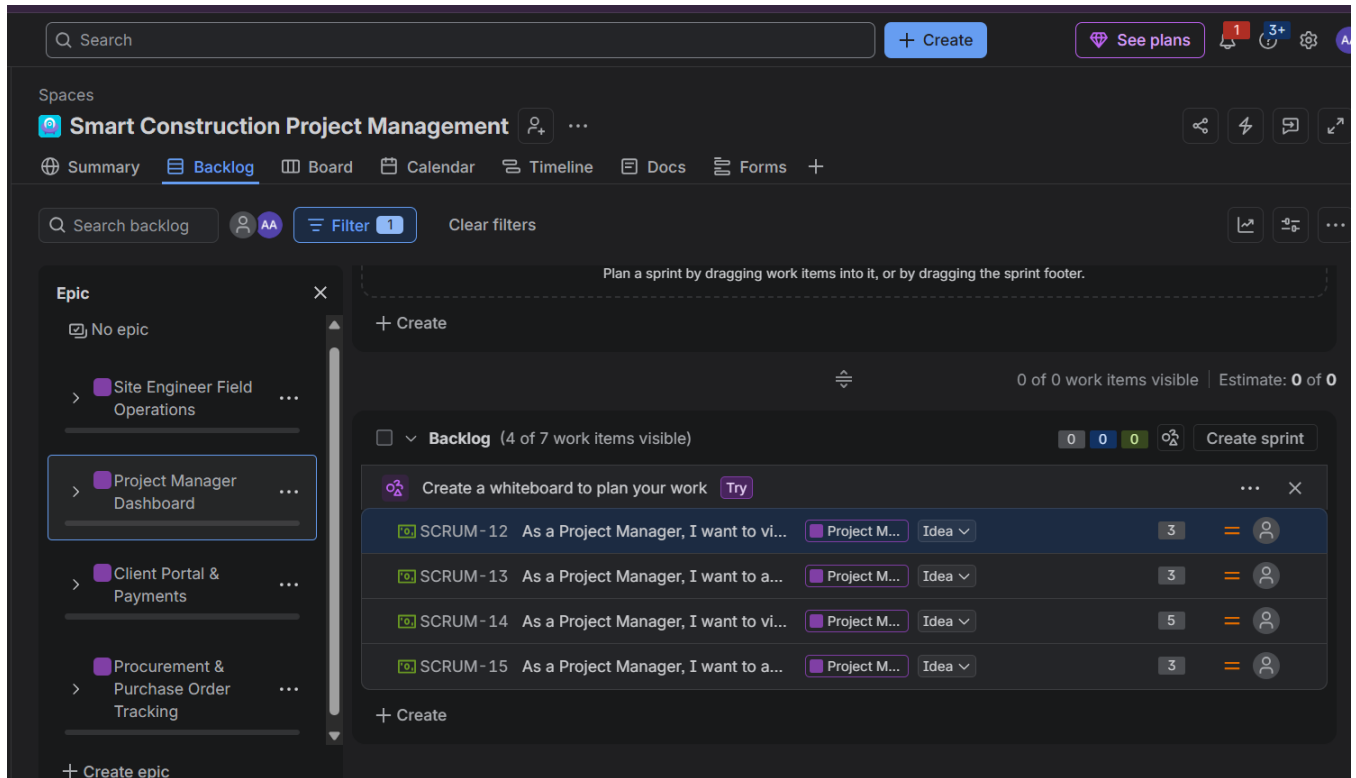
(Mobile app — progress logging, material requisitions)



- **US-1.1:** As a Site Engineer, I want to log daily construction progress with photos/notes, so that project managers have an accurate, real-time record of site status.
- **US-1.2:** As a Site Engineer, I want to update task completion status against the schedule, so that delays are visible immediately rather than at the end of the week.
- **US-1.3:** As a Site Engineer, I want to raise a material requisition request from the field, so that I don't have to wait until I'm back at the office to report shortages.

Epic 2: Project Manager Dashboard

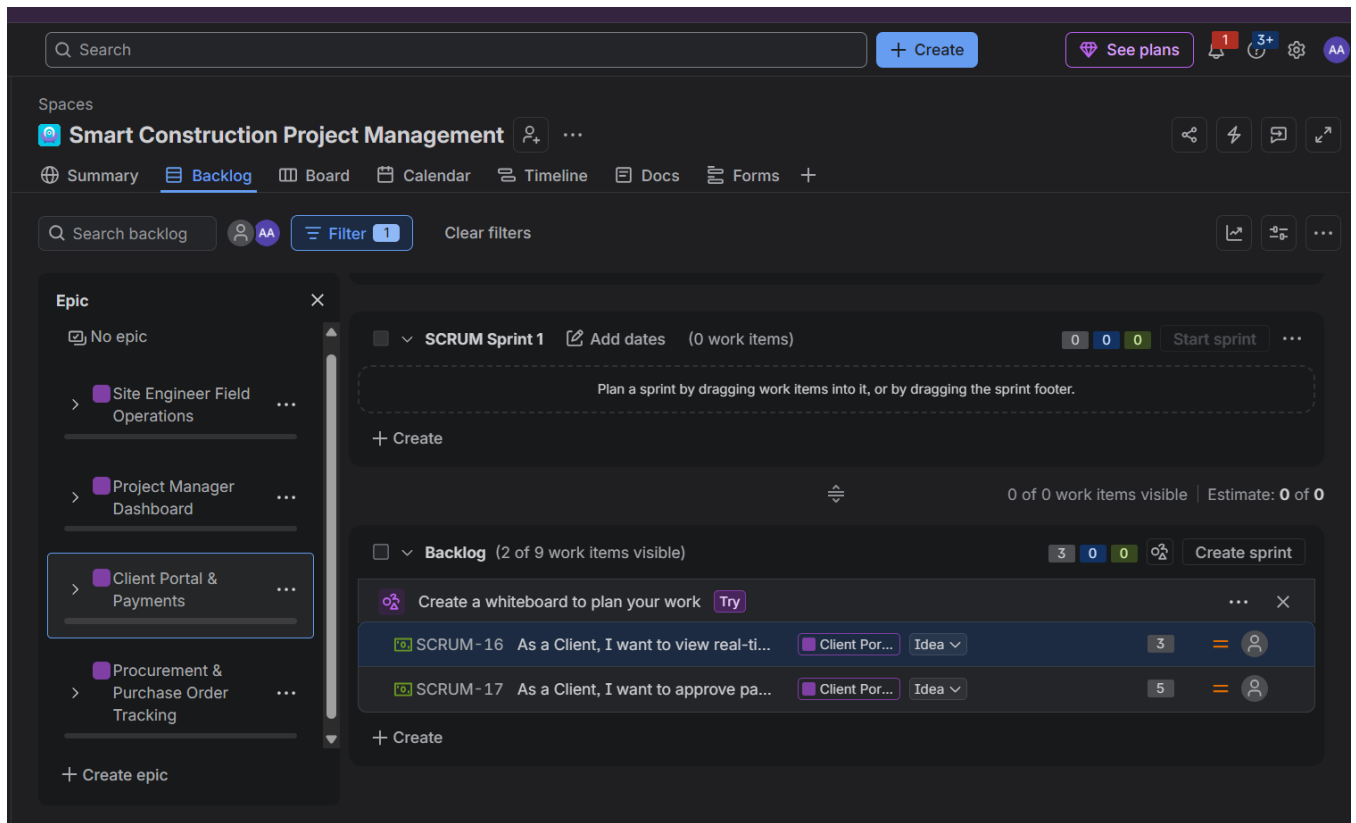
(Web dashboard — schedule tracking, approvals)



- **US-2.1:** As a Project Manager, I want to view a consolidated dashboard of all site progress updates, so that I can track schedule health across multiple projects at once.
- **US-2.2:** As a Project Manager, I want to approve or reject material requisitions, so that procurement only acts on validated requests.
- **US-2.3:** As a Project Manager, I want to view and adjust project schedules against actual progress, so that I can re-plan when tasks fall behind.
- **US-2.4:** As a Project Manager, I want to approve site-reported milestones, so that only verified progress is shared with the client.

Epic 3: Client Portal & Payments

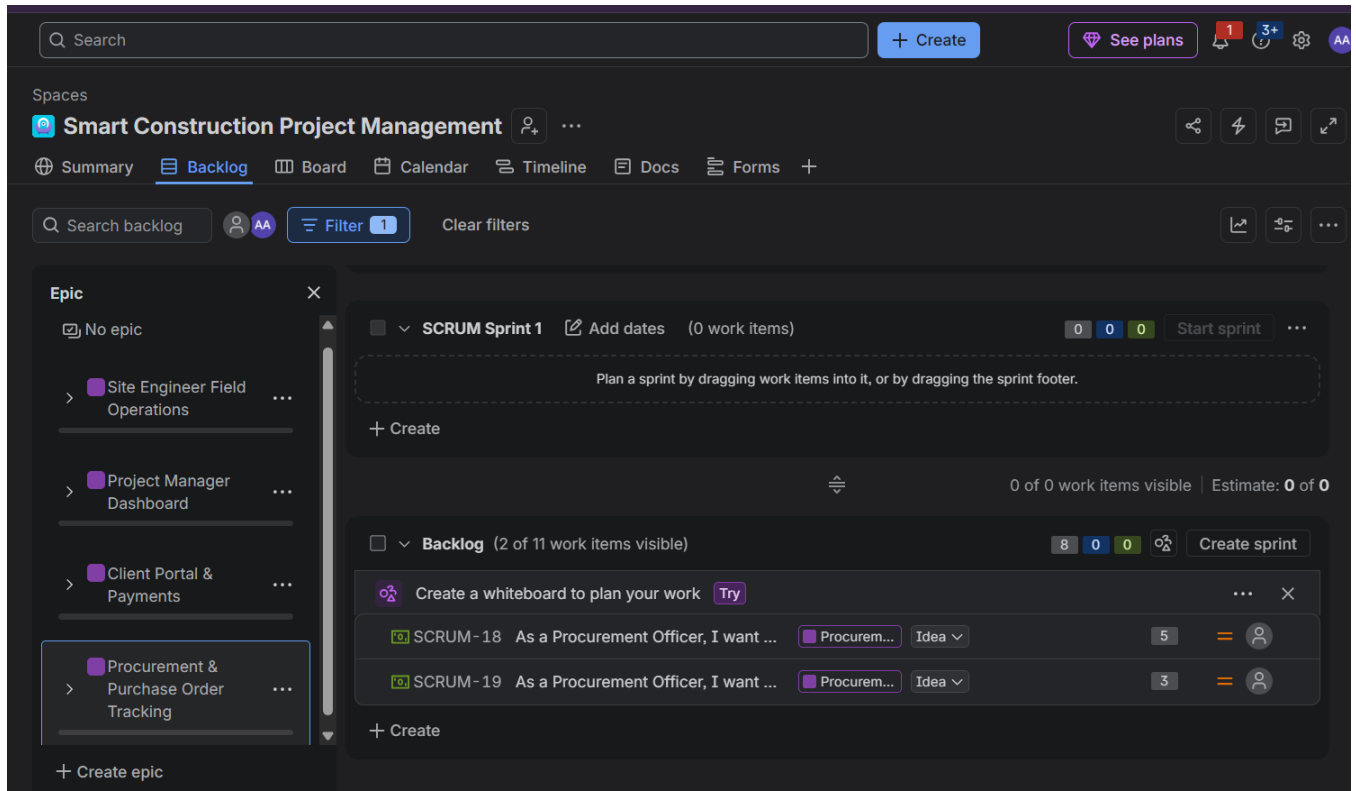
(Client portal — milestone progress + payment approval)



- **US-3.1:** As a Client, I want to view real-time milestone progress, so that I have confidence in project status without needing manual updates.
- **US-3.2:** As a Client, I want to approve payments tied to completed and verified milestones, so that payments are released only against actual work done.

Epic 4: Procurement & Purchase Order Tracking

(Procurement queue — purchase order tracking)

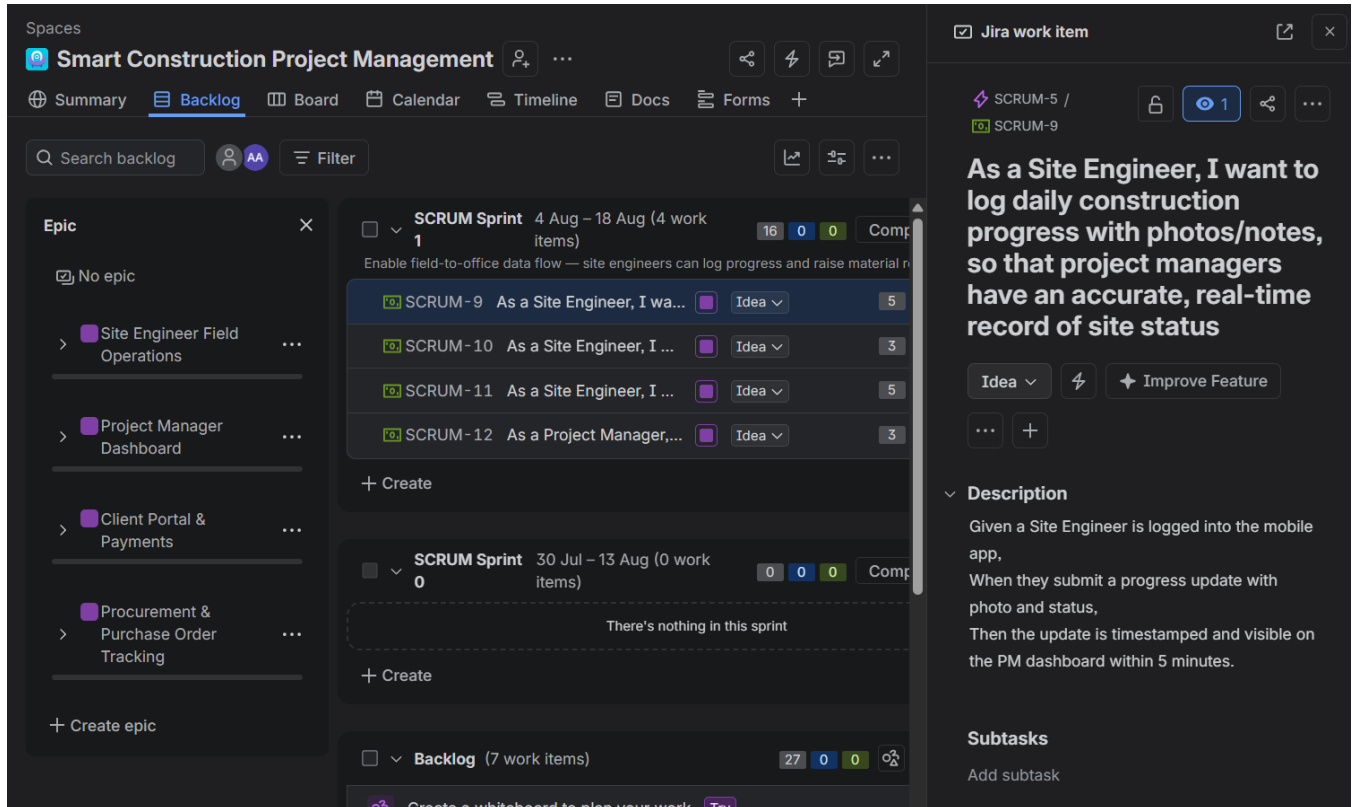


- **US-4.1:** As a Procurement Officer, I want to convert approved requisitions into purchase orders, so that material ordering starts as soon as a request is validated.
- **US-4.2:** As a Procurement Officer, I want to track purchase order delivery status, so that I can flag delays before they impact site work.

3. Acceptance Criteria (Given–When–Then)

US-1.1 — Log daily progress

- Given a Site Engineer is logged into the mobile app, When they submit a progress update with photo and status, Then the update is timestamped and visible on the PM dashboard within 5 minutes.

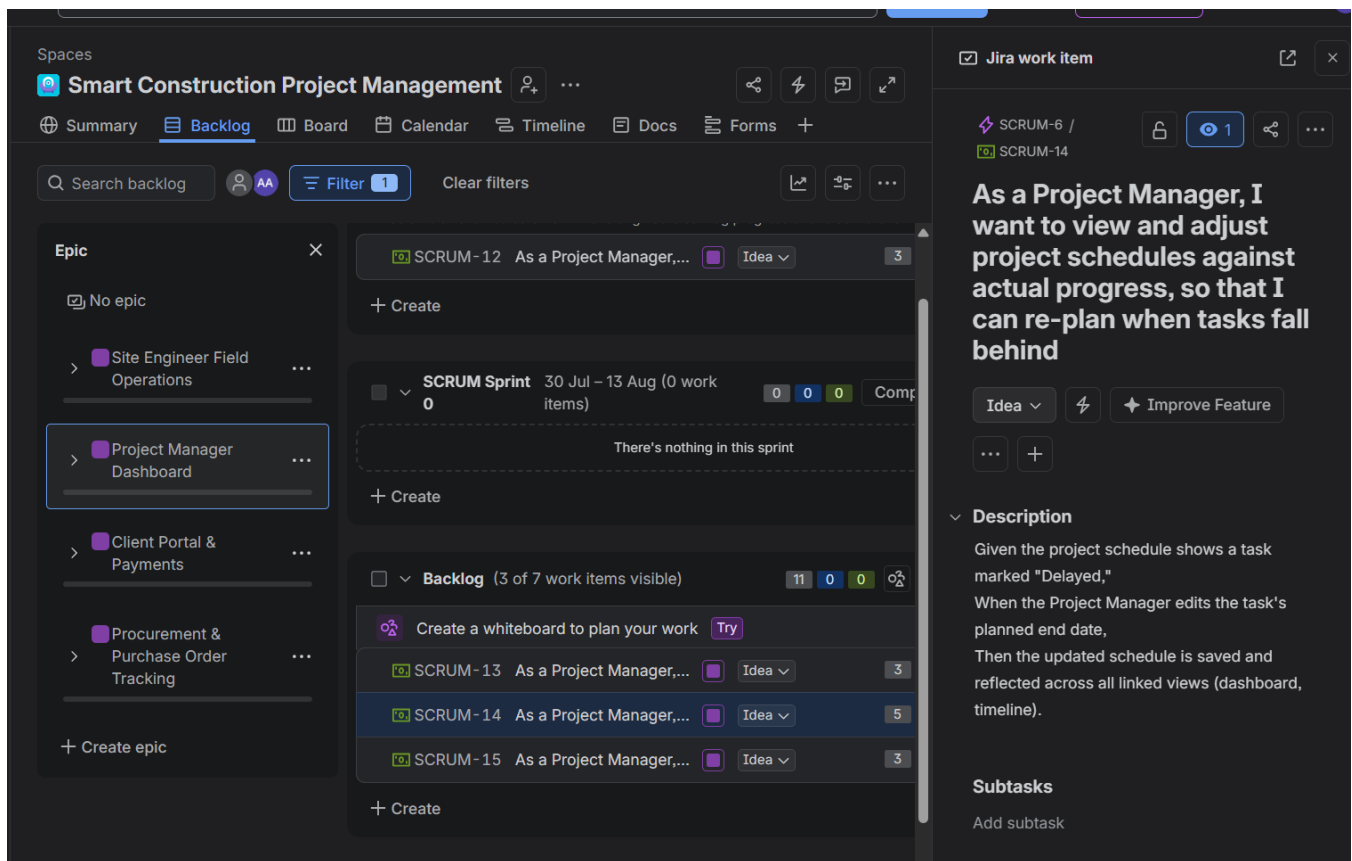


US-1.3 — Raise material requisition

- Given a Site Engineer identifies a material shortage, When they submit a requisition specifying material, quantity, and urgency, Then the request appears in the Project Manager's pending-approval queue with a unique requisition ID.

US-2.2 — Approve/reject requisition

- Given a requisition is pending, When the Project Manager approves or rejects it with a comment, Then the Site Engineer and Procurement team are notified, and the status updates in the system.

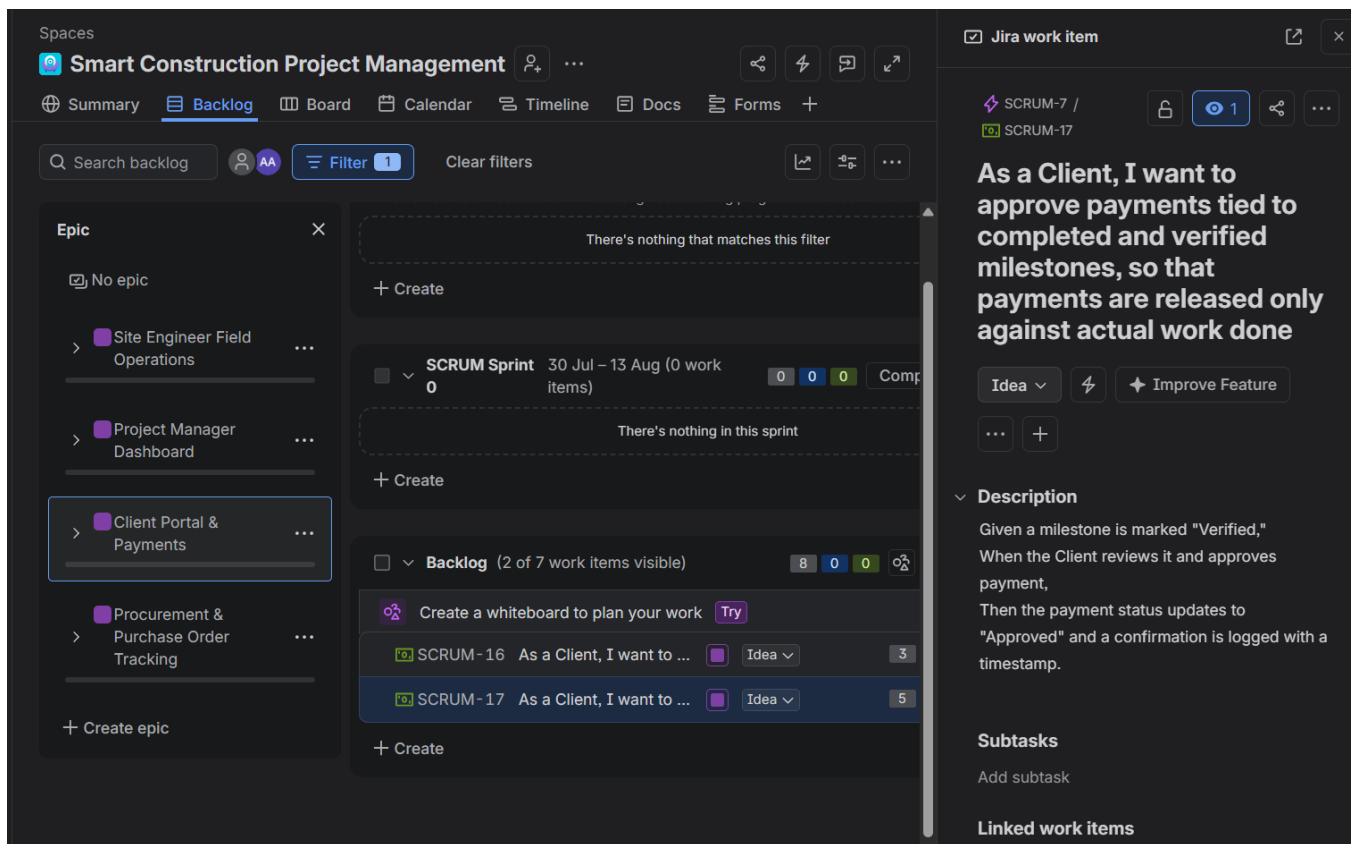


US-2.4 — Approve milestone

- Given a milestone's linked tasks are marked complete by the Site Engineer, When the Project Manager reviews and approves the milestone, Then the milestone status changes to "Verified" and becomes visible on the Client Portal.

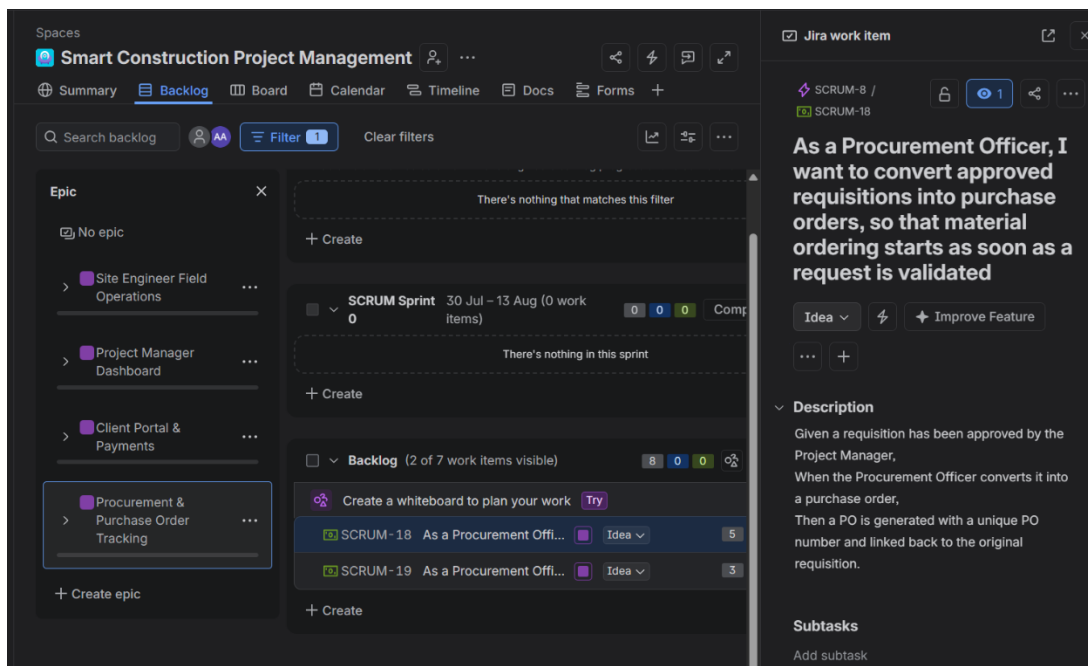
US-3.2 — Client payment approval

- Given a milestone is marked "Verified," When the Client reviews it and approves payment, Then the payment status updates to "Approved" and a confirmation is logged with a timestamp.



US-4.1 — Convert requisition to purchase order

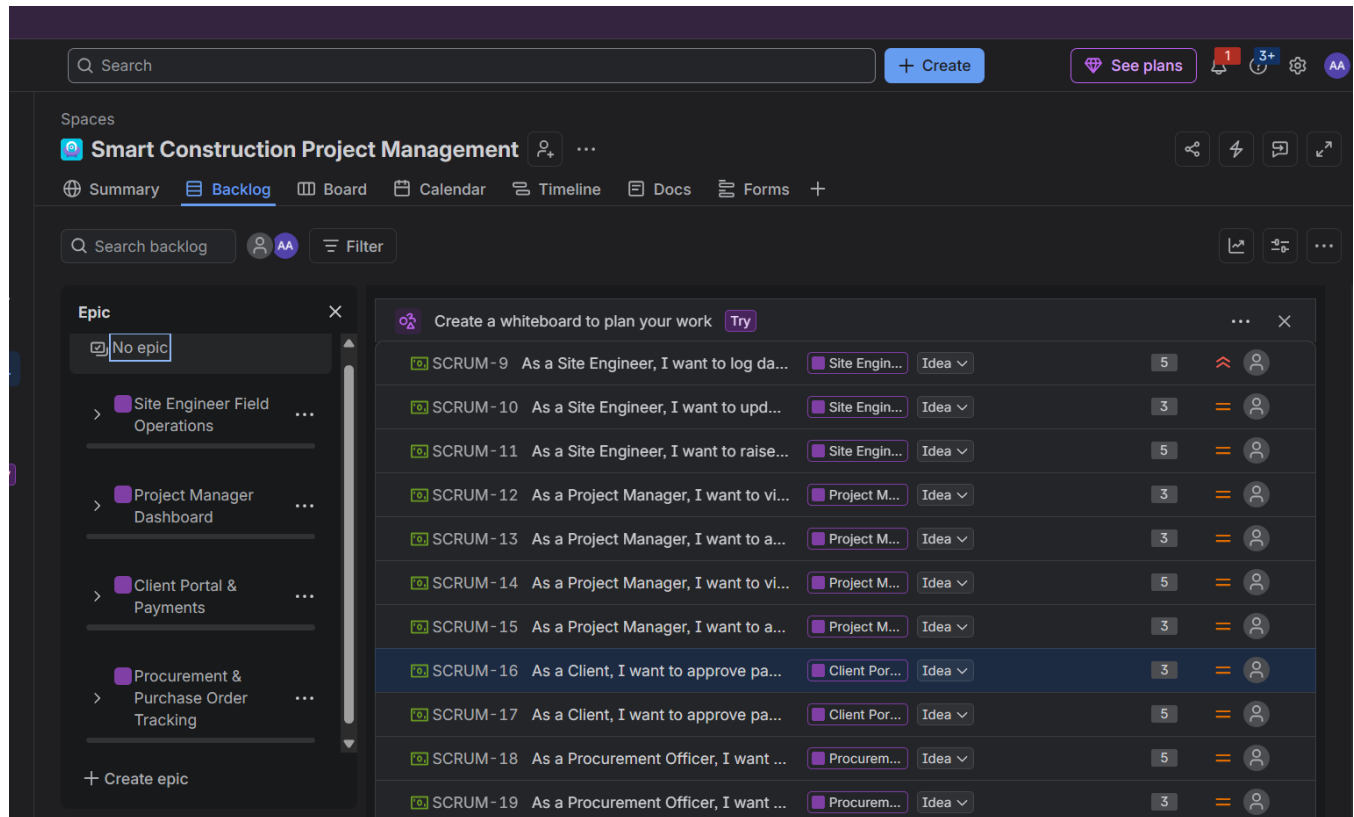
- Given a requisition has been approved by the Project Manager, When the Procurement Officer converts it into a purchase order, Then a PO is generated with a unique PO number and linked back to the original requisition.



US-4.2 — Track PO delivery status

- Given a purchase order has been issued, When the delivery status changes (e.g., Dispatched, Delayed, Delivered), Then the Procurement Officer and Site Engineer both see the updated status in real time.

Full backlog (all epics, no filter)



4. Product Backlog Prioritization

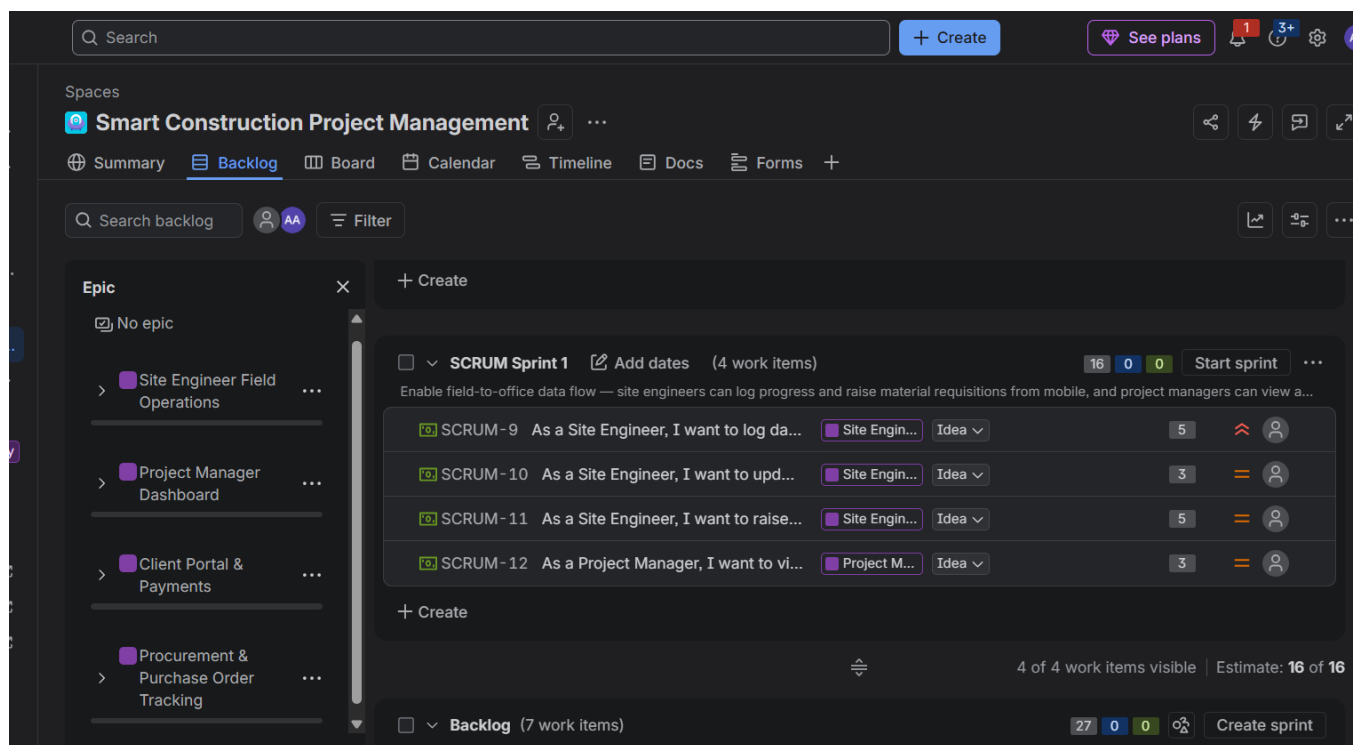
Prioritized using **MoSCoW**, weighted by business value and dependency order:

Priority	User Story	Justification
Must Have	US-1.1, US-1.2	Core data entry — nothing else works without progress data flowing in
Must Have	US-1.3, US-2.2	Directly addresses the case study's stated pain point (procurement delays)
Must Have	US-2.3, US-2.4	Schedule tracking and milestone verification are the PM's core function

Should Have	US-2.1	Dashboard aggregation builds on progress data already captured
Should Have	US-4.1, US-4.2	Depends on requisitions being approved first
Should Have	US-3.1, US-3.2	Client-facing value, but depends on verified milestones existing first

5. Sprint Backlog — Sprint 1

Selected stories (Must Have, dependency-first): US-1.1, US-1.2, US-1.3, US-2.2



User Story	Development Tasks
US-1.1	Design progress-log DB schema · Build mobile progress-entry UI · Implement photo upload · Build API endpoint · Sync to PM dashboard
US-1.2	Link tasks to schedule model · Build status-update UI component · Update schedule status on save
US-1.3	Design requisition schema · Build requisition form (mobile) · Build API endpoint · Generate unique requisition ID

US-2.2	Build PM approval queue UI (web) · Implement approve/reject action · Trigger notification on decision
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6. Story Point Estimation (Fibonacci)

User Story	Story Points	Justification
US-1.1	5	Involves UI, file/photo upload, API, and real-time sync — moderate complexity
US-1.2	3	Reuses schedule model; mainly UI + status update logic
US-1.3	5	New schema, form, API, and ID generation logic
US-2.2	3	Builds on existing requisition data; mainly UI + notification trigger
Sprint 1 Total	16	Team of 3–4 devs, 2-week sprint — realistic for this scope

(Estimated via Planning Poker consensus — points reflect complexity + effort + uncertainty, not just time.)

7. Sprint Goal & Deliverables

Sprint Goal: Enable field-to-office data flow — site engineers can log progress and raise material requisitions from mobile, and project managers can view and act on them from the web dashboard.

Expected Deliverables:

- Functional mobile progress-logging screen (site engineer)
- Functional mobile requisition-raising flow
- Web dashboard showing incoming progress logs and pending requisitions
- Approve/reject workflow for requisitions, with notification trigger
- Underlying DB schema for progress logs and requisitions

Spaces

Smart Construction Project Management

...

Summary

Backlog

Board

Calendar

Timeline

Docs

Forms

+

Q Search board

AA

Filter

Group

Complete sprint

...

Idea 4

As a Site Engineer, I want to log daily construction progress with photos/notes, so that project managers have an accurate, real-time record of site status

Parent

Site Engineer Field Operations

SCRUM-9

5

As a Site Engineer, I want to update task completion status against the schedule, so that delays are visible immediately rather than at the end of the week

Parent

Site Engineer Field Operations

SCRUM-10

3

+ Create

To Do 0

+ Create

In Progress 0

+ Create

Done 0

+ Create

Spaces

Smart Construction Project Management

...

Summary

Backlog

Board

Calendar

Timeline

Docs

Forms

+

Q Search board

AA

Filter

Group

Complete sprint

...

Idea 4

As a Site Engineer, I want to raise a material requisition request from the field, so that I don't have to wait until I'm back at the office to report shortages

Parent

Site Engineer Field Operations

SCRUM-11

5

As a Project Manager, I want to view a consolidated dashboard of all site progress updates, so that I can track schedule health across multiple projects at once

Parent

Project Manager Dashboard

SCRUM-12

3

+ Create

To Do 0

+ Create

In Progress 0

+ Create

Done 0

+ Create

Epic ↔ Figma Screen Reference

Epic	Figma Screen
Epic 1: Site Engineer Field Operations	Mobile app (site engineer)
Epic 2: Project Manager Dashboard	Web dashboard (project manager)
Epic 3: Client Portal & Payments	Client portal
Epic 4: Procurement & Purchase Order Tracking	Procurement queue